

# High-Speed I/O Design Notes

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# Chapter 1

## Signal Integrity

### 1.1 Overview

High-speed I/O channels introduce attenuation, reflection, dispersion, and inter-symbol interference (ISI).

A transmitted symbol is therefore distorted by the channel before it reaches the receiver.

The channel can be modeled as a linear time-invariant system with impulse response

$$h_{\text{ch}}(t). \tag{1.1}$$

If the transmitted pulse is

$$x_{\text{pulse}}(t), \tag{1.2}$$

then the channel pulse response is

$$p_{\text{ch}}(t) = x_{\text{pulse}}(t) * h_{\text{ch}}(t). \tag{1.3}$$

For a bandwidth-limited channel, the pulse response usually contains a large main cursor and a long post-cursor tail.

## Chapter 2

# Equalization

### 2.1 Overview

Equalization is used to compensate for channel loss and reduce inter-symbol interference.

Common equalization techniques include:

- Continuous-Time Linear Equalizer (CTLE)
- Feed-Forward Equalizer (FFE)
- Decision-Feedback Equalizer (DFE)

The CTLE operates in continuous time and compensates for frequency-dependent channel attenuation.

The DFE operates using previously detected symbols and is primarily used to cancel post-cursor ISI.

### 2.2 Continuous-Time Linear Equalizer

#### 2.2.1 CTLE Transfer Function

Consider a typical first-order CTLE transfer function

$$H_{\text{CTLE}}(s) = A_0 \frac{1 + s/\omega_z}{1 + s/\omega_p}, \quad (2.1)$$

where

$$\omega_z < \omega_p. \quad (2.2)$$

The zero appears at a lower frequency than the pole, producing high-frequency peaking. At low frequency,

$$\omega \ll \omega_z, \quad (2.3)$$

the gain is approximately

$$H(j\omega) \approx A_0. \quad (2.4)$$

At high frequency,

$$\omega \gg \omega_p, \quad (2.5)$$

the magnitude approaches

$$|H(j\omega)| \approx A_0 \frac{\omega_p}{\omega_z}. \quad (2.6)$$

Therefore, the high-frequency boost relative to the low-frequency gain is

$$G_{\text{boost}} = 20 \log_{10} \left( \frac{\omega_p}{\omega_z} \right). \quad (2.7)$$

Thus,

CTLE compensates channel high-frequency attenuation.

(2.8)

### 2.2.2 CTLE Impulse Response

Rewrite the transfer function as

$$H_{\text{CTLE}}(s) = A_0 \frac{\omega_p}{\omega_z} \frac{s + \omega_z}{s + \omega_p}. \quad (2.9)$$

Using

$$\frac{s + \omega_z}{s + \omega_p} = 1 + \frac{\omega_z - \omega_p}{s + \omega_p}, \quad (2.10)$$

we obtain

$$H_{\text{CTLE}}(s) = A_0 \frac{\omega_p}{\omega_z} \left[ 1 + \frac{\omega_z - \omega_p}{s + \omega_p} \right]. \quad (2.11)$$

Taking the inverse Laplace transform,

$$h_{\text{CTLE}}(t) = A_0 \frac{\omega_p}{\omega_z} \delta(t) + A_0 \frac{\omega_p}{\omega_z} (\omega_z - \omega_p) e^{-\omega_p t} u(t). \quad (2.12)$$

Since

$$\omega_p > \omega_z, \quad (2.13)$$

the exponential term has a negative coefficient.

Define

$$K = A_0 \frac{\omega_p}{\omega_z}, \quad (2.14)$$

and

$$C = A_0 \frac{\omega_p}{\omega_z} (\omega_p - \omega_z), \quad (2.15)$$

where

$$K > 0, \quad C > 0. \quad (2.16)$$

The CTLE impulse response therefore becomes

$$h_{\text{CTLE}}(t) = K\delta(t) - Ce^{-\omega_p t}u(t) \quad (2.17)$$

This consists of

$$\text{positive instantaneous response} + \text{negative exponentially decaying tail.} \quad (2.18)$$

### 2.2.3 Channel Pulse Response

Let the transmitted pulse be

$$x_{\text{pulse}}(t). \quad (2.19)$$

The channel pulse response is

$$p_{\text{ch}}(t) = x_{\text{pulse}}(t) * h_{\text{ch}}(t). \quad (2.20)$$

Because the channel is bandwidth limited, the received pulse is generally broadened in time.

The response therefore contains

$$\text{main cursor} + \text{pre-cursor ISI} + \text{post-cursor ISI.} \quad (2.21)$$

### 2.2.4 Pulse Response After the CTLE

After the CTLE, the new pulse response is

$$p_{\text{eq}}(t) = p_{\text{ch}}(t) * h_{\text{CTLE}}(t). \quad (2.22)$$

Using the continuous-time convolution definition,

$$p_{\text{eq}}(t) = \int_{-\infty}^{+\infty} p_{\text{ch}}(\tau) h_{\text{CTLE}}(t - \tau) d\tau. \quad (2.23)$$

Substituting

$$h_{\text{CTLE}}(t) = K\delta(t) - Ce^{-\omega_p t}u(t), \quad (2.24)$$

gives

$$p_{\text{eq}}(t) = \int_{-\infty}^{+\infty} p_{\text{ch}}(\tau) \left[ K\delta(t - \tau) - Ce^{-\omega_p(t-\tau)}u(t - \tau) \right] d\tau. \quad (2.25)$$

Separate the two terms:

$$p_{\text{eq}}(t) = K \int_{-\infty}^{+\infty} p_{\text{ch}}(\tau) \delta(t - \tau) d\tau \quad (2.26)$$

$$- C \int_{-\infty}^{+\infty} p_{\text{ch}}(\tau) e^{-\omega_p(t-\tau)} u(t - \tau) d\tau. \quad (2.27)$$

Using the sifting property,

$$\int_{-\infty}^{+\infty} p_{\text{ch}}(\tau) \delta(t - \tau) d\tau = p_{\text{ch}}(t). \quad (2.28)$$

Also,

$$u(t - \tau) = 0 \quad \text{for} \quad \tau > t. \quad (2.29)$$

Therefore,

$$p_{\text{eq}}(t) = K p_{\text{ch}}(t) - C \int_{-\infty}^t p_{\text{ch}}(\tau) e^{-\omega_p(t-\tau)} d\tau \quad (2.30)$$

This is the most important continuous-time expression for understanding CTLE equalization.

### 2.2.5 Interpretation of the Integral Term

Define

$$m(t) = \int_{-\infty}^t p_{\text{ch}}(\tau) e^{-\omega_p(t-\tau)} d\tau. \quad (2.31)$$

Then

$$p_{\text{eq}}(t) = K p_{\text{ch}}(t) - C m(t) \quad (2.32)$$

The term  $m(t)$  represents an exponentially weighted history of the channel pulse response. It can be written as

$$m(t) = p_{\text{ch}}(t) * [e^{-\omega_p t} u(t)]. \quad (2.33)$$

The corresponding transfer function is

$$H_{\text{LP}}(s) = \frac{1}{s + \omega_p}. \quad (2.34)$$

Therefore,

$$m(t) = \text{a first-order low-pass filtered version of } p_{\text{ch}}(t) \quad (2.35)$$

apart from a constant gain factor.

### 2.2.6 Low-Pass Differential Equation

Starting from

$$m(t) = \int_{-\infty}^t p_{\text{ch}}(\tau) e^{-\omega_p(t-\tau)} d\tau, \quad (2.36)$$

differentiate with respect to time.

Using Leibniz's rule,

$$\frac{dm(t)}{dt} = p_{\text{ch}}(t) + \int_{-\infty}^t p_{\text{ch}}(\tau) \frac{\partial}{\partial t} e^{-\omega_p(t-\tau)} d\tau. \quad (2.37)$$

Since

$$\frac{\partial}{\partial t} e^{-\omega_p(t-\tau)} = -\omega_p e^{-\omega_p(t-\tau)}, \quad (2.38)$$

we obtain

$$\frac{dm(t)}{dt} = p_{\text{ch}}(t) - \omega_p \int_{-\infty}^t p_{\text{ch}}(\tau) e^{-\omega_p(t-\tau)} d\tau. \quad (2.39)$$

Therefore,

$$\boxed{\frac{dm(t)}{dt} + \omega_p m(t) = p_{\text{ch}}(t)} \quad (2.40)$$

which is the standard first-order low-pass differential equation.

Taking the Laplace transform,

$$sM(s) + \omega_p M(s) = P_{\text{ch}}(s). \quad (2.41)$$

Thus,

$$\boxed{\frac{M(s)}{P_{\text{ch}}(s)} = \frac{1}{s + \omega_p}} \quad (2.42)$$

which confirms the low-pass behavior.

### 2.2.7 Why the Main Cursor Is Preserved and the Tail Is Reduced

The equalized response is

$$p_{\text{eq}}(t) = Kp_{\text{ch}}(t) - Cm(t). \quad (2.43)$$

Consider a rapidly changing portion of the channel pulse response.

During a fast transition,

$$p_{\text{ch}}(t) \quad (2.44)$$

can change rapidly, whereas the low-pass quantity

$$m(t) \quad (2.45)$$

cannot respond instantaneously.

Therefore,

$$m(t) \quad (2.46)$$

is relatively small compared with the instantaneous term, giving

$$p_{\text{eq}}(t) \approx Kp_{\text{ch}}(t). \quad (2.47)$$

Thus, the rapidly changing main cursor is preserved or emphasized.

For a slowly varying component, however,  $m(t)$  has enough time to follow the waveform.

If

$$p_{\text{ch}}(t) = P \quad (2.48)$$

for a sufficiently long time, then

$$m(t) = P \int_0^\infty e^{-\omega_p u} du \quad (2.49)$$

$$= \frac{P}{\omega_p}. \quad (2.50)$$

The output becomes

$$p_{\text{eq}} = KP - C \frac{P}{\omega_p}. \quad (2.51)$$

Therefore, slowly varying components experience stronger cancellation.  
The CTLE can therefore be interpreted as

$$\boxed{\text{CTLE output} = \text{instantaneous signal} - \text{low-pass version of the signal.}} \quad (2.52)$$

Equivalently,

$$\boxed{\text{high-frequency emphasis} = \text{signal} - \text{slow component.}} \quad (2.53)$$

The CTLE does not explicitly identify the main cursor or post-cursor.  
Instead,

$$\boxed{\text{main cursor} \rightarrow \text{more fast-changing content,}} \quad (2.54)$$

while

$$\boxed{\text{long tail} \rightarrow \text{more slowly varying content.}} \quad (2.55)$$

### 2.2.8 Over-Equalization

The CTLE can also overcompensate the channel.

From

$$p_{\text{eq}}(t) = Kp_{\text{ch}}(t) - Cm(t), \quad (2.56)$$

if

$$Cm(t) > Kp_{\text{ch}}(t), \quad (2.57)$$

then

$$\boxed{p_{\text{eq}}(t) < 0.} \quad (2.58)$$

Thus, an originally positive post-cursor can become negative.  
For example, suppose

$$Kp_{\text{ch}}(1\text{UI}) = 0.28, \quad (2.59)$$

and

$$Cm(1UI) = 0.35. \quad (2.60)$$

Then

$$p_{eq}(1UI) = 0.28 - 0.35 = -0.07. \quad (2.61)$$

Hence,

$$\boxed{\text{excessive CTLE boost} \rightarrow \text{over-equalization} \rightarrow \text{opposite-sign post-cursor}.} \quad (2.62)$$

Therefore, maximizing CTLE peaking is not the design objective.

Instead, the objective is to maximize eye opening and minimize the total sampled ISI.

### 2.2.9 From Continuous-Time Pulse Response to Discrete Cursors

After equalization,

$$p_{eq}(t) = p_{ch}(t) * h_{CTLE}(t). \quad (2.63)$$

Let the symbol period be

$$T_{UI}. \quad (2.64)$$

Choose a sampling phase

$$t_s. \quad (2.65)$$

The sampled coefficients are

$$\boxed{c[k] = p_{eq}(t_s + kT_{UI}).} \quad (2.66)$$

The main cursor is

$$\boxed{c[0] = p_{eq}(t_s).} \quad (2.67)$$

The first post-cursor is

$$\boxed{c[1] = p_{eq}(t_s + T_{UI}).} \quad (2.68)$$

The second post-cursor is

$$\boxed{c[2] = p_{eq}(t_s + 2T_{UI}).} \quad (2.69)$$

Similarly, the first pre-cursor is

$$\boxed{c[-1] = p_{eq}(t_s - T_{UI}).} \quad (2.70)$$

Therefore,

$$\boxed{c[k] = [p_{ch} * h_{CTLE}](t_s + kT_{UI}).} \quad (2.71)$$

These are the discrete-time coefficients seen by the sampler and the DFE.

### 2.2.10 Why Sampled Pulse Response Becomes a Discrete Channel

Let the transmitted symbol sequence be

$$a[n]. \quad (2.72)$$

The transmitted waveform is

$$x(t) = \sum_{n=-\infty}^{+\infty} a[n] x_{\text{pulse}}(t - nT_{\text{UI}}). \quad (2.73)$$

After the channel and CTLE,

$$y(t) = \sum_{n=-\infty}^{+\infty} a[n] p_{\text{eq}}(t - nT_{\text{UI}}). \quad (2.74)$$

Sample at

$$t = mT_{\text{UI}} + t_s. \quad (2.75)$$

Then

$$y[m] = \sum_n a[n] p_{\text{eq}}(t_s + (m - n)T_{\text{UI}}). \quad (2.76)$$

Using

$$c[k] = p_{\text{eq}}(t_s + kT_{\text{UI}}), \quad (2.77)$$

we obtain

$$\boxed{y[m] = \sum_n a[n] c[m - n]}. \quad (2.78)$$

Equivalently,

$$y[m] = a[m]c[0] + a[m - 1]c[1] + a[m - 2]c[2] + \cdots. \quad (2.79)$$

Thus,

$$\boxed{c[0] = \text{main cursor}}, \quad (2.80)$$

and

$$\boxed{c[1], c[2], \dots = \text{post-cursor ISI coefficients.}} \quad (2.81)$$

### 2.2.11 Sampling Before and After Convolution

For two continuous-time signals

$$x(t) \tag{2.82}$$

and

$$h(t), \tag{2.83}$$

the continuous-time convolution is

$$y(t) = (x * h)(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau. \tag{2.84}$$

Sampling after convolution gives

$$y[n] = y(nT_s) = \int_{-\infty}^{+\infty} x(\tau)h(nT_s - \tau)d\tau. \tag{2.85}$$

If the signals are sampled first,

$$x[n] = x(nT_s), \tag{2.86}$$

and

$$h[n] = h(nT_s), \tag{2.87}$$

then the discrete convolution is

$$y_d[n] = \sum_{k=-\infty}^{+\infty} x(kT_s)h((n - k)T_s). \tag{2.88}$$

In general,

$$\text{sample}\{x * h\} \neq \text{sample}\{x\} * \text{sample}\{h\}. \tag{2.89}$$

The reason is that continuous-time convolution uses every value of the waveform between sampling instants.

The exact sampled continuous-time convolution can be written as

$$y(nT_s) = \sum_k \int_{kT_s}^{(k+1)T_s} x(\tau)h(nT_s - \tau)d\tau. \tag{2.90}$$

If  $T_s$  is sufficiently small, each integral can be approximated by

$$\int_{kT_s}^{(k+1)T_s} x(\tau)h(nT_s - \tau)d\tau \approx T_s x(kT_s)h((n - k)T_s). \tag{2.91}$$

Therefore,

$$y(nT_s) \approx T_s \sum_k x(kT_s)h((n - k)T_s). \tag{2.92}$$

Thus, high-rate discrete convolution can approximate continuous-time convolution.

However, sampling only once per UI before convolution generally loses important intra-UI waveform information.

### 2.2.12 Recommended Numerical Procedure

Given a measured or simulated channel pulse response  $p_{\text{ch}}(t)$ , a practical numerical flow is:

1. Obtain the CTLE impulse response

$$h_{\text{CTLE}}(t). \quad (2.93)$$

2. Choose a fine time step

$$\Delta t \ll T_{\text{UI}}. \quad (2.94)$$

3. Sample the channel pulse response

$$p_{\text{ch}}[n] = p_{\text{ch}}(n\Delta t). \quad (2.95)$$

4. Sample the CTLE impulse response

$$h_{\text{CTLE}}[n] = h_{\text{CTLE}}(n\Delta t). \quad (2.96)$$

5. Perform numerical convolution

$$p_{\text{eq}}[n] \approx \Delta t (p_{\text{ch}} * h_{\text{CTLE}})[n]. \quad (2.97)$$

6. Select the sampling phase  $t_s$ .

7. Sample once every UI

$$c[k] = p_{\text{eq}}(t_s + kT_{\text{UI}}). \quad (2.98)$$

8. Extract

$$c[0], c[1], c[2], \dots \quad (2.99)$$

as the main cursor and post-cursor coefficients.

## 2.3 Summary

The CTLE equalization process can be summarized as

$$\boxed{p_{\text{ch}}(t) \xrightarrow{*h_{\text{CTLE}}(t)} p_{\text{eq}}(t) \xrightarrow{\text{sample every UI}} c[k].} \quad (2.100)$$

The CTLE impulse response can be written as

$$\boxed{h_{\text{CTLE}}(t) = K\delta(t) - Ce^{-\omega_p t}u(t).} \quad (2.101)$$

The equalized pulse response is

$$\boxed{p_{\text{eq}}(t) = Kp_{\text{ch}}(t) - C \int_{-\infty}^t p_{\text{ch}}(\tau)e^{-\omega_p(t-\tau)}d\tau.} \quad (2.102)$$

The first term represents the instantaneous component, while the second term subtracts an exponentially weighted low-pass version of the waveform.

Therefore,

$$\boxed{\text{fast-changing waveform components} \rightarrow \text{relatively emphasized},} \quad (2.103)$$

while

$$\boxed{\text{slowly varying channel tails} \rightarrow \text{relatively suppressed}.} \quad (2.104)$$

Excessive peaking may cause

$$\boxed{\text{over-equalization} \rightarrow \text{negative post-cursors}.} \quad (2.105)$$

The sampled equalized pulse response

$$\boxed{c[k] = p_{\text{eq}}(t_s + kT_{\text{UI}})} \quad (2.106)$$

provides the discrete-time cursor coefficients used for ISI and DFE analysis.

## Chapter 3

# Decision-Feedback Equalization

This chapter will discuss DFE architecture, feedback timing, critical-path constraints, and PAM-3 DFE implementation.

## Chapter 4

# PAM-3 Receiver Architecture

This chapter will discuss PAM-3 slicers, upper and lower thresholds, VREFH/VREFL, receiver offset calibration, and tri-level decision logic.

## Chapter 5

# Clocking and Timing

This chapter will discuss sampling clocks, forwarded clocks, clock phase control, duty-cycle correction, and timing margin.

## Chapter 6

# GDDR7 Receiver Design

This chapter will discuss GDDR7 receiver architecture, training, VREF calibration, CTLE, DFE, slicers, and timing.

# Notes

Additional design notes, references, and derivations can be added here.